

INTRODUCTION

AI-Based Toxic Content and Potential Cyberbullying Detection

Social media platforms allow users to communicate and share information rapidly, but they can also be used to spread hateful, toxic, and abusive content. **Cyberbullying** is a significant concern because harmful messages can reach a large audience and may be difficult to identify manually when dealing with thousands of online posts.

Machine learning provides a way to automatically analyse large amounts of textual data and identify patterns associated with hateful or non-hateful content. In this project, Twitter text is processed and converted into numerical representations using **TF-IDF**, allowing machine-learning algorithms to analyse the text mathematically.

The project implements four classification algorithms: **Logistic Regression, SVM, Naive Bayes, and Random Forest**. These models use the same processed textual features but employ different learning approaches. Their predictions are evaluated on an unseen test dataset using standard classification metrics. Furthermore, ensemble models are created by combining classifiers with different learning mechanisms. The purpose is to investigate whether combining their predictions can reduce individual model errors and improve the robustness of toxic-content classification.

Overall, the project demonstrates a complete machine-learning pipeline—from **text preprocessing and feature extraction to classification, evaluation, and ensemble modelling**—for detecting potentially harmful content in social-media text.

Abstract

The rapid growth of social media has increased the occurrence of toxic content and cyberbullying, creating a need for automated detection systems. This project, **AI-Based Toxic Content and Potential Cyberbullying Detection**, applies machine-learning techniques to classify text as hateful or non-hateful content. A labelled dataset of **31,962 tweets** was used. The text was preprocessed by removing usernames, URLs, punctuation, special characters and stopwords, followed by stemming. **TF-IDF (Term Frequency–Inverse Document Frequency)** was then used to convert the processed text into numerical features. Four machine-learning algorithms — **Logistic Regression, Support Vector Machine (SVM), Naive Bayes, and Random Forest**—were trained and evaluated. Their performance was measured using accuracy, precision, recall, and F1-score. In addition, ensemble models were developed by combining classifiers to investigate whether combining different learning approaches could provide more robust predictions. The final models were also compared using technical parameters such as training time, prediction time, learned parameters, model size, non-zero learned values, and prediction confidence.